**Introduction to Software Engineering**

# Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.




Software Engineering Department
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# Software Requirements Specification

## Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

# 1 Member Contribution Assessment

PA2 contribution allocation confirmed on 02 Nov 2026: 25% for each member, totaling 100%. Evidence links identify assigned work; completion and review status are recorded separately.

| ID | Name | Role(s) | Contribution (%) | Evidence |
|---|---|---|---|---|
| 24127169 | Trương Lê Trung Hiếu | PM / technical coordination | 25% | DID-21, DID-43, DID-45, DID-56-58 |
| 24127130 | Trần Minh Tiến | Business Analyst | 25% | DID-40-42, DID-44, DID-46, DID-49, DID-55 |
| 24127020 | Nguyễn Thành Đạt | Designer / UX | 25% | DID-50-52, Figma |
| 24127035 | Võ Đình Cao Minh Hào | Tester / requirements QA | 25% | DID-47-48, DID-53-54 |
| Total | | | 100% | |

# 2 Introduction

## 2.1 Purpose and Scope

*DiDung is a responsive web service for accessing air-quality information and short-term PM2.5 forecasts for supported locations in Ho Chi Minh City. The MVP includes data ingestion, an AQI/PM2.5 dashboard, forecast views, account preferences, configurable thresholds, and in-app notifications. Flood and high-tide prediction are feasibility-study topics only, not committed MVP functions. Target readers: users, BA, designer, developer and tester; course staff are external readers only. PA1 is the latest scope baseline where it supersedes PA0. TA is not a project participant or approver. Current BA feedback on open requirements and decisions remains pending.*

## 2.2 Glossary

| Term | Definition |
|---|---|
| AQI | Air Quality Index: a numerical index derived from pollutant data under a specified standard. |
| PM2.5 | Fine particulate matter with aerodynamic diameter at most 2.5 micrometres. |
| MVP | Minimum Viable Product: minimum approved release scope. |
| FR / NFR | Functional / Non-functional Requirement. |
| MoSCoW | Must, Should, Could, Won't for this release. |
| Freshness | Time elapsed since the observation or model output timestamp. |
| Forecast | Estimated future pollutant values with stated horizon and provenance. |
| SRS | Software Requirements Specification. |
| UC / AC | Use Case / Acceptance Criteria. |
| RTM | Requirements Traceability Matrix. |
| Stale data | Data older than the team-approved freshness threshold (threshold pending). |

# 3 Problem Statement

*Residents and commuters in Ho Chi Minh City need accessible, understandable information about current air quality and near-term exposure risks. Existing environmental feeds vary in geographic coverage, observation versus modeled provenance, refresh intervals and completeness. DiDung aims to aggregate available PM2.5 and relevant weather information, communicate AQI with source/timestamp, provide short-term PM2.5 forecasts and support user-selected locations and alert preferences. The intended system is a responsive Vietnamese web application using the proposed Next.js UI/API, Supabase PostgreSQL/Auth and Vercel hosting. Constraints include public API quotas, incomplete or stale observations, model uncertainty, limited course time and privacy/security. It must not imply street-level measurement or medical certainty. Facebook-derived flood/tide risk is a separate feasibility study and will not be advertised as an MVP forecast without a documented scope change and verification.*

# 4 Stakeholders

| # | Stakeholder | Description | Interests / concerns |
|---|---|---|---|
| 1 | Course staff (external reader) | May read submitted coursework; does not participate in project decisions or review. | Clarity and traceability in submitted work; no project approval role. |
| 2 | Resident / visitor | Views air quality in supported locations. | Current readings, reliable source/time labels and clear missing-data states. |
| 3 | Commuter | Checks selected places and near-term forecasts. | Accessible comparisons and clearly labeled uncertainty. |
| 4 | Registered user | Sets tracked locations and alert thresholds. | Account security, correct preferences and in-app notices. |
| 5 | External data provider | Supplies permitted pollutant/weather data. | Proper attribution, quota compliance and responsible usage. |
| 6 | DiDung project team | BA, PM, designer, developer and tester. | Feasible requirements, review evidence, implementable design and testability. |

# 5 Requirements

## 5.1 Functional Requirements

| ID | Requirement | Feature | Priority | Source |
|---|---|---|---|---|
| FR-01 | *The system shall retrieve available PM2.5 and weather inputs from configured permitted sources, recording source, location, unit and timestamp.* | F01 | Must | PA1 F01; data source study DID-43 |
| FR-02 | The system shall store validated observations or modeled inputs and flag missing, invalid or stale records rather than silently substituting values. | F01 | Must | PA1 F01; BA assumption pending |
| FR-03 | The system shall display available AQI/PM2.5 for a selected supported location with source, timestamp and data status. | F07 | Must | PA1 F07; residents |
| FR-04 | The system shall display location-related AQI/PM2.5 on a map and provide available historical values for the selected location/time range. | F07 | Must | PA1 F07; commuters |
| FR-05 | The system shall present a short-term PM2.5 forecast with forecast horizon, issue time and distinction from observed values. | F02 | Must | PA1 F02; residents |
| FR-06 | The system shall register and authenticate users using a configured identity service and prevent unauthorized access to account settings. | F05 | Must | PA1 F05; security requirement |
| FR-07 | The system shall let authenticated users save, edit and delete tracked locations and personal alert thresholds. | F05 | Must | PA1 F05; registered users |
| FR-08 | The system shall evaluate eligible air-quality data against a user's active alert threshold and create an in-app notice when configured trigger conditions occur. | F06 | Must | PA1 F06; registered users |
| FR-09 | The system shall display a user's notifications and allow them to mark individual notices as read. | F06 | Must | PA1 F06; registered users |

| ID | Requirement | Feature | Priority | Source |
|---|---|---|---|---|
| FR-10 | The system shall present an explicit error, empty or stale state when requested air-quality data are unavailable or outdated. | F07 | Must | PA1 F07; error-handling analysis |
| FR-11 | The system should present available baseline-versus-ML forecast evaluation results with clearly identified metrics and datasets. | F03 | Should | PA1 F03 |
| FR-12 | The system could display general, non-medical interpretation guidance corresponding to the selected AQI standard. | F08 | Could | PA1 F08 |
| FR-13 | The system should provide system administrators with a monitoring view of pipeline run status, stale-data warnings, deployment health and forecast-error summaries. | F09 | Should | PA1 F09 (Should; latest scope; PA0 nice-to-have is historical) |
| FR-14 | The system shall provide a publicly accessible landing page that describes DiDung's purpose and main capabilities, states that product features require sign-in, and provides navigation to registration and sign-in. | F07, F05 | Must | PA1 F07 – Dashboard; PA1 F05 – Accounts and regional alerts. Public landing page and authenticated product access. |

## 5.2 Non-Functional Requirements

| ID | Category | Requirement (measurable) | How it will be verified |
|---|---|---|---|
| NFR-01 | Performance | Proposed target: under 50 simulated concurrent users, cached dashboard GET requests shall complete in <=2s at P95, excluding third-party data refresh. | Run repeatable load test; report environment, payload, P95 and failure rate. |
| NFR-02 | Security | Protected account/preferences APIs shall return 401/403 for unauthenticated/unauthorized requests in 100% of enumerated security test cases. | Integration/auth tests on protected routes; review RLS policies. |

| ID | Category | Requirement (measurable) | How it will be verified |
|---|---|---|---|
| NFR-03 | Usability | At least 4 of 5 representative users shall complete location selection and AQI inspection unaided within 60 seconds in a moderated test (proposed target). | Usability task test with recorded times, environment and consent. |
| NFR-04 | Accessibility | Primary text and control states shall meet WCAG 2.2 AA contrast targets: 4.5:1 normal text, 3:1 large text and required non-text indicators where applicable. | Contrast audit of design and implemented screens; keyboard testing on implementation. |
| NFR-05 | Reliability | When a provider returns an error, the UI shall display a distinct unavailable/stale state, never a newly fabricated measurement, in 100% of scripted failure tests. | Inject API 4xx/5xx/timeouts and inspect screen/API output. |
| NFR-06 | Data integrity | All displayed pollutant values shall retain source and timestamp metadata in 100% of synthetic ingestion tests. | Schema/check ingestion tests for null/units/timezone/provenance. |
| NFR-07 | Responsive design | For viewport widths 390px and 1440px, main user flows shall show no horizontal scrolling caused by layout overflow in the defined screen set. | Browser responsive audit; document browsers and exceptions. |
| NFR-08 | Privacy | User location preferences and notification records shall not be readable by another account in any of the account-isolation test cases. | Two-user RLS/API isolation test, no credentials/tokens in evidence. |

## 5.3 Domain Requirements and Business Rules

| ID | Rule | Source (policy, law, standard) |
|---|---|---|
| BR-01 | The team shall declare one chosen AQI scale and calculation/interpretation source; it must not mix scale categories without labeling. | Approved AQI standard/policy - pending selection |
| BR-02 | Observations and forecasts must be labeled distinctly; every datum must expose source and observation/issue time. | PA1 data provenance; team decision pending |

| ID | Rule | Source (policy, law, standard) |
|---|---|---|
| BR-03 | Staleness and forecast horizon thresholds must be defined per source/feature before acceptance testing. | Provider documentation; proposed team rule |
| BR-04 | Missing measurements must never be silently replaced with zero or a current reading from an unrelated location. | Domain data quality rule |
| BR-05 | Threshold-based notifications shall use the configured comparator, avoid unintended duplicates and record trigger time. | Stakeholder needs; team rule pending |
| BR-06 | Only the account owner may access or change saved preferences and in-app notification history. | Account security/privacy policy |
| BR-07 | Flood/tide forecasts from social posts remain outside MVP until feasibility and formal change-control approval. | PA1 Scope A; PA1 is the latest baseline where it supersedes PA0. |

## 5.4 Acceptance Criteria

*AC-F01-01: Given a configured, permitted source returns a well-formed pollutant or weather record with its source, location, unit and timestamp, when ingestion validates and stores the record, then the stored record retains those fields and is marked valid; a modeled value is not relabeled as observed.*

*AC-F01-02: Given the source input is absent or fails a required field/type/unit validation, when ingestion processes the input, then it is marked rejected/invalid, no new valid measurement is created from it, and the rejection reason is recorded.*

*AC-F01-03: Given one or more configured permitted sources return available PM2.5/weather records for a supported location, when the scheduled ingestion run executes, then it retrieves the available inputs and records the run outcome and source/location for each retrieved record. Exact provider coverage and update cadence remain subject to DID-43/source validation.*

*AC-F01-04: Given an input timestamp is older than the approved freshness threshold for that source, when ingestion validation evaluates the input, then the record is flagged stale and is not silently relabeled as current or replaced by a value from another location. The approved source-specific threshold remains unspecified (BR-03).*

*AC-F01-05: Given a configured source API returns HTTP 4xx/5xx or times out, when a scheduled ingestion attempt calls that source, then the attempt is marked failed with source and attempt time, no new valid measurement is created from that failed response, and previously stored valid data are not overwritten by fabricated data. Retry/backoff limits remain implementation design for PA3/PA4.*

*AC-F02-01: Given a supported location has an available PM2.5 forecast whose target horizon is from 1 through 6 hours, when a Web User opens the forecast, then the estimate, forecast horizon, issue time, source/provenance and forecast label are shown distinctly from observations.*

*AC-F02-02: Given no forecast output is available, including because the configured forecast/provider API is unavailable, returns an error or times out, when a Web User opens or refreshes the forecast, then the UI shows an explicit unavailable state and does not create or display a fabricated forecast curve/value.*

*AC-F05-01: Given the configured identity service recognizes valid credentials, when the Web User signs in, then authentication succeeds and the protected preferences view is available only to that authenticated account.*

*AC-F05-02: Given credentials are invalid or rejected by the configured identity service, when the Web User submits sign-in, then access is denied with a non-secret error response and no protected preferences are disclosed.*

*AC-F05-03: Given an authenticated user selects a supported location and provides a valid threshold under the currently configured rule, when the user saves the preference, then it persists under that user's account and is returned as that user's saved preference. The AQI scale/comparator must be configured before acceptance (BR-01/BR-05).*

*AC-F05-04: Given a user submits a registration request with details accepted by the configured identity service, when registration completes, then an account is created by that service and can authenticate through the same service. Provider-specific validation/verification remains as defined by the selected provider.*

*AC-F05-05: Given a requester is unauthenticated, or an authenticated requester does not own the target account/preferences, when the requester reads or changes protected settings, saved locations or alert thresholds, then the request returns an unauthorized/forbidden result (401/403 as applicable), no protected data are disclosed, and the owner's stored preference remains unchanged.*

*AC-F05-06: Given an authenticated user owns an existing saved location or alert threshold, when the user edits or deletes that preference, then the persisted record reflects the edit or is absent after deletion, and unrelated account preferences remain unchanged.*

*AC-F06-01: Given an active user threshold has a saved comparator, a supported location has an eligible fresh value, and the value crosses the saved threshold rule, when the threshold evaluator processes that event, then exactly one in-app notice is recorded for the event with its location and trigger time. The comparator and equality boundary are not chosen here; they remain a BR-05 decision.*

*AC-F06-02: Given an event has already produced an in-app notice and the same event identity is submitted again, when duplicate suppression processes the replay, then the total number of notices for that event remains exactly one. The event identity and suppression window remain BR-05 decisions.*

*AC-F06-03: Given an authenticated user owns an unread notification, when the user marks it read, then its persisted read status is `read` and remains `read` after the notification center is reopened.*

*AC-F06-04: Given an authenticated user owns one or more in-app notifications, when that user opens the notification center, then only that user's notifications are displayed with their stored read/unread state.*

*AC-F06-05: Given a fresh eligible value does not satisfy the user's saved comparator/threshold rule, when the evaluator processes the value, then zero notifications are created for that evaluation. The equal-to-threshold case follows the BR-05 boundary decision and remains pending until that decision is recorded.*

*AC-F06-06: Given an input is flagged stale or invalid under the configured validation rules, when the threshold evaluator processes it, then zero notifications are created from that input and the data quality status remains available for inspection. The eligibility rule requires BA/Tester confirmation before acceptance sign-off.*

*AC-F06-07: Given authenticated user B does not own a notification belonging to user A, when B attempts to read or change A's notification/read status, then the request returns 401/403 as applicable, exposes no notification content, and leaves A's notification unchanged. This aligns with NFR-02/NFR-08 and BR-06.*

*AC-F07-01: Given a supported location has a valid fresh AQI/PM2.5 value, when a Web User selects that location, then the value, source, timestamp and observed/modeled data status are shown.*

*AC-F07-02: Given a requested value is older than the approved freshness threshold, when a Web User selects the location, then the UI labels the value stale, displays its true timestamp and does not present it as current. The freshness threshold remains BR-03/TBD.*

*AC-F07-03: Given a selected location is unsupported or has no valid data, when the Web User requests its AQI/PM2.5, then the UI displays an explicit no-data/unsupported state and no fabricated value or unrelated-location value.*

*AC-F07-04: Given historical/map values exist for a supported location and requested time range, when a Web User opens the map/history view, then displayed values correspond only to that location and time range and retain the source/timestamp provenance of the underlying records.*

*AC-F07-05: Given the dashboard provider API returns HTTP 4xx/5xx or times out, when a Web User refreshes the dashboard, then the UI shows an understandable unavailable/stale state, does not fabricate a new measurement, and exposes a retry action.*

Additional draft criteria — FR-11–FR-13

Proposed for review; priorities and release scope remain unchanged. These criteria are not test results.

AC-F03-01 : Given comparable baseline and ML evaluation results, when the evaluation view is opened, then show each model, metric, unit, dataset/evaluation period and result. The metric and comparison protocol require team confirmation.

AC-F03-02 : Given a model result is missing or evaluation conditions differ, when results are viewed, then identify the missing/non-comparable data without substituting zero or claiming model superiority.

AC-F08-01 : Given a valid AQI under the standard approved in BR-01, when interpretation guidance is viewed, then shows the corresponding category and approved general guidance, clearly identified as non-medical.

AC-F08-02 : Given AQI is missing/invalid or the applicable standard is unknown, when guidance is viewed, then state that classification is unavailable and do not present guidance as though air quality were Good.

AC-F09-01 : Given an authorized System Administrator and available monitoring data, when the monitoring view is opened, then show pipeline status, stale-data warnings, deployment health and forecast-error summaries with source/time; mark missing data unavailable.

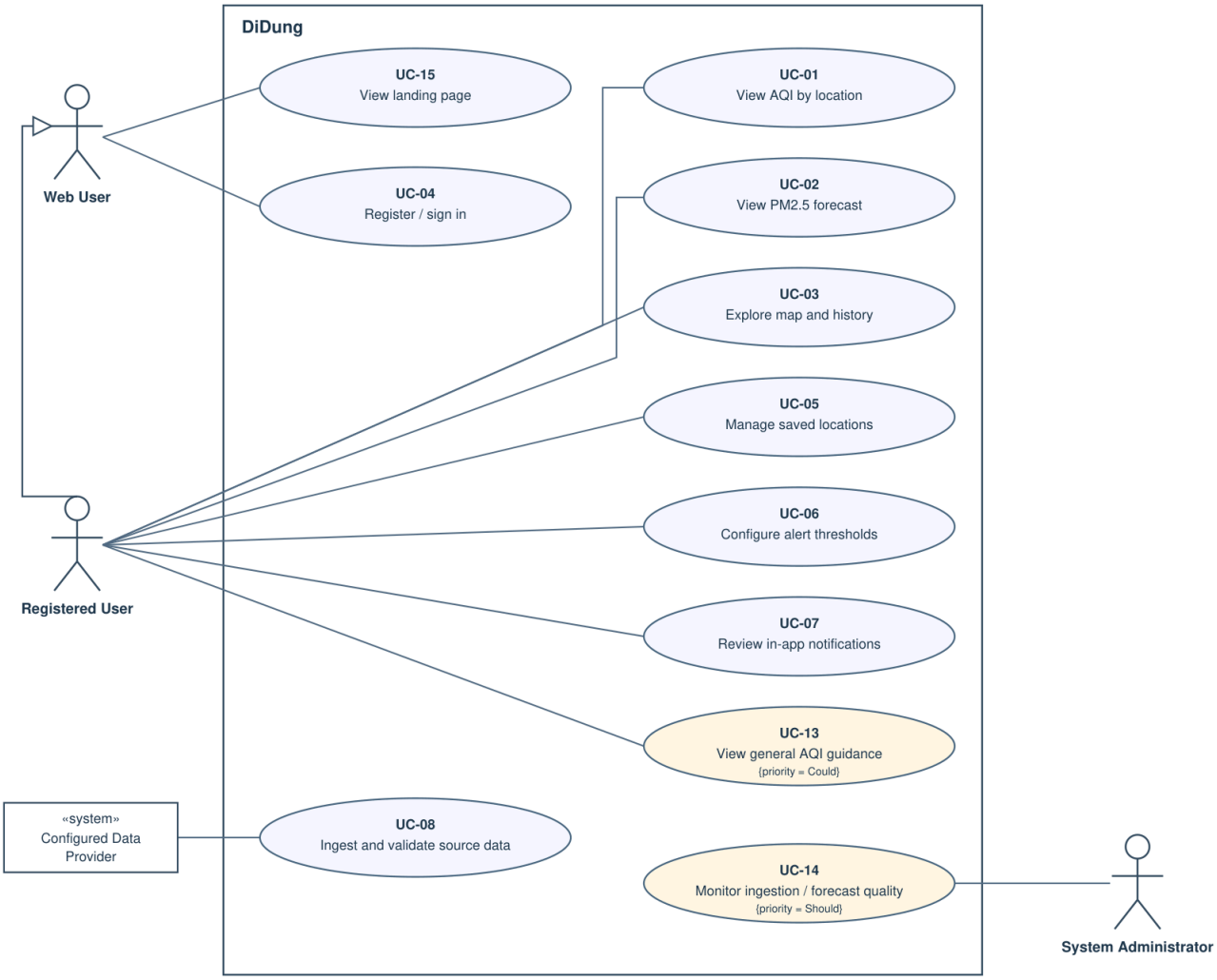
AC-F09-02 : Given an unauthenticated or non-admin requester, when protected monitoring data are requested, then deny access with 401/403 as applicable and disclose no protected monitoring content.

AC-FR14-01: Given a visitor is not signed in, when they open the landing page, then the system displays DiDung's purpose, main capabilities, the sign-in requirement for product features, and working registration/sign-in navigation.

# 6 Use Case Model

## 6.1 Use Case Diagram

**DiDung — UML Use Case Diagram**




Protected product use cases require a valid authenticated session; monitoring also requires administrator authorization.
UC-09/10: incorporated behavior. UC-11: internal workflow. UC-12: actor unresolved; retained in scope register.

## 6.2 Actors

| Actor | Description |
|---|---|
| Web User | General website visitor role. Can view the public landing page and enter registration/sign-in. Live product features require authentication; landing demo content is not access to protected live data. |
| Registered User | Specializes Web User. With a valid authenticated session, accesses AQI/PM2.5, forecasts, map/history, guidance, saved-location requirements, thresholds and in-app notices. |
| Configured Data Provider | External system supplying permitted air-quality/weather inputs. The owner reports the production map uses Open-Meteo/CAMS model data; the exact configured model/domain, returned grid, API payload and ingestion behavior remain pending runtime validation. |
| System Administrator | System Administrator reviews pipeline run status, stale-data warnings, deployment health and forecast-error summaries (PA1 F09, Should). |

## 6.3 Use Case List

| ID | Use case | Primary actor | Priority | Related FR |
|---|---|---|---|---|
| UC-01 | View AQI by location | Registered User | Must | FR-03, FR-10 |
| UC-02 | View PM2.5 forecast | Registered User | Must | FR-05, FR-10 |
| UC-03 | Explore map and history | Registered User | Must | FR-04, FR-10 |
| UC-04 | Register / sign in | Web User | Must | FR-06 |
| UC-05 | Manage saved locations | Registered User | Must | FR-07 |
| UC-06 | Configure alert thresholds | Registered User | Must | FR-07 |
| UC-07 | Review in-app notifications | Registered User | Must | FR-09 |
| UC-08 | Ingest and validate source data | None - automatic process; external provider is supporting actor | Must | FR-01, FR-02 |
| UC-09 | Show source, time and provenance | Registered User | Must | FR-03, FR-04, FR-05; BR-02 |

| ID | Use case | Primary actor | Priority | Related FR |
|---|---|---|---|---|
| UC-10 | Show stale / unavailable state | Registered User | Must | FR-10 |
| UC-11 | Evaluate threshold and create in-app notice | None - automatic evaluation; Registered User is recipient | Must | FR-08 |
| UC-12 | View forecast evaluation results | Didn't have ML yet | Should | FR-11 |
| UC-13 | Provide general AQI guidance | Registered User | Could | FR-12 |
| UC-14 | Monitor ingestion / forecast quality | System Administrator | Should (PA1 F09) | FR-13 |
| UC-15 | View landing page | Web User | Must | FR-14 |

# 7 Use Case Specifications

Use-case review — 6 November 2026. System boundary: DiDung application, internal scheduled jobs and application persistence; external identity/data/map services are supporting actors only when they participate in that interaction. Supabase Auth is the external identity service; Supabase database/storage is treated as infrastructure, not a separate user-goal actor. User-facing product views require an authenticated Registered User; only landing/authentication entry is public. Static landing demonstrations do not establish access to live protected data.

Repository alignment is based on source inspection, not runtime acceptance. UC-09/10 are incorporated display/exception behavior; UC-08/11 are internal workflows. Existing identifiers are retained for traceability. Requirements not yet implemented remain requirements, with gaps explicitly recorded rather than silently removed.

Key evidence: src/stores/auth-store.ts; src/lib/auth-navigation.ts; src/app/app/page.tsx; src/hooks/use-map-canvas.ts; src/hooks/use-account.ts; src/app/api/air-quality/{current,history,forecast}/route.ts; src/app/api/sync/{air-quality-model,weather}/route.ts; src/app/api/jobs/evaluate-alerts/route.ts in https://github.com/entidy06/DiDung-Air-quality-assessment-service (reviewed source snapshot c27081e75b52e9073fb76d97e11a4f5110959323; fetched files carry their own blob SHAs).

Remaining scope: complete saved-location account CRUD is not established; forecast issue-time and 1–6h conformance remain gaps; full ingestion validation/rejection logging is not established. Admin route currently guards access but AdminShell is a placeholder, not FR-13 monitoring implementation. UC-12's actor remains unresolved. The document currently contains detailed tables UC-01–11 only; optional UC-12–14 and landing UC-15 still need their own specifications. UC-15 should trace to the observable landing-page FR-14, not an untestable promise of user understanding.

## 7.1 UC-01 - View AQI by Location

| *Use case ID* | UC-01 |
|---|---|
| *Use case name* | View AQI by location |

| *Use case ID* | UC-01 |
|---|---|
| *Brief description* | Registered User selects a supported location and sees available AQI/PM2.5 together with provenance and data state. |
| *Primary actor* | Registered User |
| *Secondary actors* | None in this viewing interaction. The system reads stored model data; Open-Meteo participates in UC-08, not in the dashboard request. Supabase persistence is treated as infrastructure within the DiDung service boundary. |
| *Priority* | Must |
| *Related requirements* | FR-03, FR-10; BR-01 to BR-04 |
| *Trigger* | Registered User opens the dashboard or selects a location. |
| *Pre-conditions* | The user has a valid authenticated session; the application is reachable and supported locations are configured. |
| *Post-conditions* | Available stored model values for the selected location are shown, or a truthful unavailable/stale state is shown. Display does not establish physical-station measurements or successful live ingestion. |
| *Main flow* | *1. Registered User opens the authenticated dashboard.*<br>*2. System presents supported locations.*<br>*3. User selects a location.*<br>*4. System retrieves stored current air-quality model data for that location through the protected application API.*<br>*5. System prepares available source, valid-time and fetched-time metadata without treating them as interchangeable.*<br>*6. System displays available AQI/PM2.5 with modeled-data labels and the applicable data state.* |
| *Alternative flows* | *A1 (step 3): user searches for another supported location; continue at step 4.*<br>*A2 (step 5): source is stale; label stale and show original timestamp.* |
| *Exception flows* | *E1 (step 4): retrieval fails or no current record exists; show an unavailable/no-data state without inventing readings.*<br>*E2 (step 3): an unsupported location is selected; explain supported coverage.*<br>*E3 (entry or retrieval): session is absent or expired; redirect the web flow to sign-in and withhold protected data; protected API returns 401.* |

| *Use case ID* | UC-01 |
|---|---|
| *Business rules* | BR-01, BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-01, NFR-04, NFR-05, NFR-07 |

## 7.2 UC-02 - View PM2.5 Forecast

| *Use case ID* | UC-02 |
|---|---|
| *Use case name* | View PM2.5 Forecast |
| *Brief description* | A Registered User views available short-term PM2.5 forecasts for a selected supported location, with the forecast period, provenance and available time metadata, or an explicit unavailable state. |
| *Primary actor* | Registered User |
| *Secondary actors* | None in this viewing interaction. Forecast data are read from stored Open-Meteo model output; no separate ML service or live data-provider call participates in this request. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-05, FR-10 |
| *Trigger* | User opens forecast for a supported location. |
| *Pre-conditions* | Registered User has a valid authenticated session and a supported location is selected. |
| *Post-conditions* | Available future model values are shown as forecasts, or unavailability is disclosed. Requirement for PM2.5 at 1–6 hours with issue time remains a conformance target; the current API defaults to 24 hours and caps requests at 72 hours, and does not return a model issue-time field. |
| *Main flow* | *1. Registered User opens the protected forecast view.*<br>*2. User selects a location and an available horizon control, if provided by the UI.*<br>*3. System retrieves stored future model records for that location.*<br>*4. System identifies modeled origin, target times and available provenance; missing issue-time metadata are not invented.*<br>*5. System displays the forecast and its available metadata or an explicit unavailable state.* |
| *Alternative flows* | *A1 (step 2): user changes location or an available horizon option; repeat retrieval at step 3. API support for a horizon is not evidence that a corresponding UI control exists.* |

| *Use case ID* | UC-02 |
|---|---|
| *Exception flows* | *E1 (step 3): no output is available or retrieval fails; explain unavailability without fabricating a curve and permit retry.*<br>*E2 (entry/retrieval): no valid session; route to sign-in and withhold protected content (API 401).* |
| *Business rules* | BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-01, NFR-05, NFR-07 |

## 7.3 UC-03 - Explore Map and History

| *Use case ID* | UC-03 |
|---|---|
| *Use case name* | Explore Map and History |
| *Brief description* | A Registered User explores air-quality information on the map and available historical values for a selected supported location and time range, with source and time labels. |
| *Primary actor* | Registered User |
| *Secondary actors* | None for retrieving stored map/history records. Mapbox is a supporting external map-rendering service only for the map subflow; it does not supply AQI, PM2.5 or history records. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-04, FR-10 |
| *Trigger* | User opens map or history view. |
| *Pre-conditions* | Registered User has a valid authenticated session; supported locations are configured. |
| *Post-conditions* | Available modeled map/history values are displayed for the selected location/time range, or no-data/error status is shown. Past model valid times are not represented as independently observed station history. |
| *Main flow* | *1. Open map/history.*<br>*2. Select location/time range.*<br>*3. Request available records.* |

| *Use case ID* | UC-03 |
|---|---|
| | *4. Label time, source and data coverage.*<br>*5. Render records on the map/chart.* |
| *Alternative flows* | *A1 (at main step 2): Registered User changes the requested time range or supported region.*<br>*A1.1 Repeat the request from main step 3.* |
| *Exception flows* | *E1 (step 3): no records exist for the requested location/range; show no-data and allow another selection.*<br>*E2 (step 3): retrieval fails; show error/retry without fabricated values.*<br>*E3 (entry/retrieval): no valid session; route to sign-in and withhold protected content (API 401).*<br>*E4 (map rendering): map service/token/network is unavailable; report the map failure without implying that stored air-quality retrieval necessarily failed.* |
| *Business rules* | BR-02, BR-04 |
| *Non-functional constraints* | NFR-04, NFR-07 |

## 7.4 UC-04 - Sign in or Register

| *Use case ID* | UC-04 |
|---|---|
| *Use case name* | Sign in or Register |
| *Brief description* | A Web User registers or signs in through the configured identity service to access account-specific features; unsuccessful requests do not disclose protected data. |
| *Primary actor* | Web User |
| *Secondary actors* | Supabase Auth — external identity service supporting email/password authentication, registration and session establishment. Google Identity Provider — additional supporting actor only for the Google OAuth alternative, mediated by Supabase Auth. Application stores, callbacks and UI components are not actors. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-06 |

| *Use case ID* | UC-04 |
|---|---|
| *Trigger* | A signed-out visitor selects sign-in/register from the public landing page, or attempts to open a protected application route. |
| *Pre-conditions* | The visitor is signed out. Supabase Auth is configured. Email sign-up/verification and Google OAuth availability follow the deployed provider configuration. |
| *Post-conditions* | On successful authentication, a valid session is established and the user reaches a safe authorized destination: /app for an ordinary user, /admin for an administrator, or an allowed requested path. Registration requiring email confirmation does not by itself establish a session. |
| *Main flow* | *1. Web User opens the authentication page.*<br>*2. User enters email and password.*<br>*3. User submits sign-in.*<br>*4. System requests email/password authentication through Supabase Auth.*<br>*5. On success, system establishes the session and determines the authorized role/destination.*<br>*6. System opens the allowed requested destination or role default.* |
| *Alternative flows* | *A1 (step 2): user selects registration; system submits sign-up through Supabase Auth. If confirmation is required and no session is returned, show the next required verification action without claiming sign-in. Continue to step 5 only when a valid session exists.*<br>*A2 (step 2): user selects Google sign-in; Supabase Auth mediates consent/authentication with Google. On callback, exchange a valid code for a session and continue to step 5.*<br>*A3 (entry): user already has a valid session; route to the authorized application destination instead of requiring repeat sign-in.* |
| *Exception flows* | *E1 (step 4): invalid credentials, rejected details or provider/network failure; show a safe failure and establish no new authenticated access.*<br>*E2 (OAuth/callback): consent is denied or the code/state is missing, expired, reused or invalid; return to authentication with an actionable safe error.*<br>*E3 (step 6): requested redirect is unsafe or outside the user's role; use the authorized role default rather than accepting the unsafe target.* |
| *Business rules* | BR-06 |

| *Use case ID* | UC-04 |
|---|---|
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.5 UC-05 - Manage Saved Locations

| *Use case ID* | UC-05 |
|---|---|
| *Use case name* | Manage Saved Locations |
| *Brief description* | A Registered User adds, views, changes or removes supported locations saved under their own account. |
| *Primary actor* | Registered User |
| *Secondary actors* | None. Account-owned saved-location management is a DiDung responsibility; its database and internal APIs are not secondary actors. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-07 |
| *Trigger* | User adds/edits/removes a followed location. |
| *Pre-conditions* | User is authenticated. |
| *Post-conditions* | Required outcome: an authorized account-owned saved-location change persists and the list reflects it, or failure is reported without success. Repository review has not established the complete saved-location CRUD flow; selected location in localStorage is not evidence of account-owned saved-location persistence. |
| *Main flow* | *1. Open account preferences.*<br>*2. View saved locations.*<br>*3. Choose add/edit/delete.*<br>*4. Validate supported location.*<br>*5. Save changes and confirm updated list.* |
| *Alternative flows* | *A1 (at main step 3): Registered User edits an existing saved location.*<br>*A1.1 Continue with validation at main step 4 and persist the accepted change at step 5.* |

| *Use case ID* | UC-05 |
|---|---|
| | *A2 (at main step 3): Registered User deletes an existing saved location.*<br>*A2.1 Persist the authorized deletion and show the updated list at main step 5.* |
| *Exception flows* | *E1 (at main step 4): the location is unsupported.*<br>*E1.1 Show the validation issue and return to main step 3.*<br>*E2 (at main step 5): persistence fails.*<br>*E2.1 Show the failure, preserve the previously saved data and do not claim success.*<br>*E2.2 Return to the saved-location view.* |
| *Business rules* | BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.6 UC-06 - Configure Alert Threshold

| *Use case ID* | UC-06 |
|---|---|
| *Use case name* | Configure Alert Threshold |
| *Brief description* | A Registered User creates or updates a location-specific alert threshold and manages its active status for subsequent automatic evaluation. |
| *Primary actor* | Registered User |
| *Secondary actors* | None. Threshold APIs, database tables and the internal evaluator are system components, not secondary actors. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-07 |
| *Trigger* | User configures a personal notification threshold. |
| *Pre-conditions* | User has a valid authenticated session. Supported location/metric definitions are available. Approved threshold/input rules are required for acceptance; current code configuration alone does not approve BR-05. |

| *Use case ID* | UC-06 |
|---|---|
| *Post-conditions* | Threshold is stored, used by eligible evaluations, or rejected as invalid. |
| *Main flow* | *1. Registered User opens alert settings.*<br>*2. User chooses a supported location and metric (AQI or PM2.5) for a new rule.*<br>*3. User enters a threshold value.*<br>*4. System validates the supplied fields; report any unsupported input.*<br>*5. System persists the rule under the authenticated account.*<br>*6. System confirms the saved rule. Subsequent automatic evaluation is the separate UC-11 workflow, not a step that must execute during configuration.* |
| *Alternative flows* | *A1 (step 3): edit an existing rule's threshold value and persist the authorized update.*<br>*A2 (step 3): toggle an existing rule's active status and confirm only after persistence succeeds.*<br>*A3 (step 3): delete an owned threshold and refresh the list after success.*<br>*Current item PATCH supports threshold_value and is_active; changing an existing rule's location/metric is not established by that route.* |
| *Exception flows* | *E1 (at main step 4): the threshold is invalid under the approved rule.*<br>*E1.1 Show an actionable validation message and return to main step 3.*<br>*E2 (at main step 5): storage fails.*<br>*E2.1 Show the failure without claiming the settings were saved.*<br>*E2.2 Return to alert settings.* |
| *Business rules* | BR-05, BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.7 UC-07 - Review In-App Notifications

| *Use case ID* | UC-07 |
|---|---|
| *Use case name* | Review In-App Notifications |

| *Use case ID* | UC-07 |
|---|---|
| *Brief description* | A Registered User views their own in-app notifications and marks individual notices as read, with persisted read status and account isolation. |
| *Primary actor* | Registered User |
| *Secondary actors* | None. In-app notification retrieval/read updates use DiDung APIs and persistence. No email, SMS, push provider or real-time delivery service is established for this use case. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-09 |
| *Trigger* | User wishes to inspect generated alerts. |
| *Pre-conditions* | User is signed in; notice history may be empty. |
| *Post-conditions* | Authorized notice list shown; read state may be updated. |
| *Main flow* | *1. Registered User opens the notification center.*<br>*2. System retrieves that account's notifications and unread count.*<br>*3. System displays the stored notice details, created time and read/unread state.*<br>*4. User chooses an unread notice's mark-read action.*<br>*5. System submits the account-scoped read-state update.*<br>*6. On success, system confirms the persisted state and refreshes the displayed notices/count.* |
| *Alternative flows* | *A1 (at main step 2): there are no notices for the account.*<br>*A1.1 Show an empty-state message.*<br>*A1.2 The use case ends without modifying stored notices.* |
| *Exception flows* | *E1 (steps 2 or 5): session absent/expired; deny protected access (API 401) and route the web flow to sign-in.*<br>*E2 (step 5): notice is not found or does not belong to the requester; the current API returns 404 without disclosing another account's notice.*<br>*E3 (steps 2 or 5): retrieval/update fails; report failure, retain truthful UI state and do not claim a successful read update.* |
| *Business rules* | BR-05, BR-06 |
| *Non-functional constraints* | NFR-02, NFR-08 |

## 7.8 UC-08 - Ingest and Validate Source Data

| *Use case ID* | UC-08 |
|---|---|
| *Use case name* | Ingest and Validate Source Data |
| *Brief description* | The system retrieves permitted pollutant/weather inputs on the configured schedule, validates the required data and stores accepted records with provenance while recording failures. |
| *Primary actor* | None - internal scheduled process |
| *Secondary actors* | Open-Meteo Air Quality API and Open-Meteo Weather API — configured external data providers. Scheduler, adapters, API routes and persistence are inside the chosen DiDung service boundary and are not actors. |
| *Priority* | Must (draft Scope A) |
| *Related requirements* | FR-01, FR-02 |
| *Trigger* | A configured pg_cron/pg_net invocation calls the relevant sync route with the cron authorization header. Deployment scheduling is configuration-dependent; source inspection does not prove a successful production run. |
| *Pre-conditions* | Active locations exist, source endpoints are configured, server-side database credentials are available and the invocation supplies the expected cron secret. |
| *Post-conditions* | Valid records are stored with provenance; malformed inputs flagged. |
| *Main flow* | *1. System validates authorization for the scheduled sync invocation.*<br>*2. System loads active location coordinates.*<br>*3. System requests Open-Meteo data in coordinate batches and parses the returned hourly series.*<br>*4. System associates each record with location, valid time and modeled origin; preserve missing values without fabricating zero readings.*<br>*5. System upserts available air-quality/weather records using their defined conflict keys.*<br>*6. System attempts to record the run status and returns success/failure for that invocation.* |

| *Use case ID* | UC-08 |
|---|---|
| *Alternative flows* | *A1 (step 3): more locations exist than fit in one request; process coordinate batches (current helper batch size: 50), not provider page-token pagination.*<br>*A2 (step 5): an existing location/time key is returned again; update the stored record through upsert rather than create a duplicate.*<br>*A3 (step 2): no active locations; no new measurement rows are created.* |
| *Exception flows* | *E1 (step 1): cron authorization is missing/incorrect; reject with 401 before ingestion.*<br>*E2 (steps 2–5): database/provider/request/parsing failure; return failure and attempt to record the sync outcome. Previously completed writes are not claimed to be rolled back.*<br>*E3 (step 4): required record validation fails; the requirement is to reject/flag invalid data, but full schema/freshness validation and per-record rejection logging are not established in the current sync routes. Retain this as an implementation gap, not a verified behavior.*<br>*No automatic retry/backoff guarantee is inferred from the current helper.* |
| *Business rules* | BR-02, BR-03, BR-04 |
| *Non-functional constraints* | NFR-05, NFR-06 |

## 7.9 UC-09 - Show Source, Time and Provenance

| Use case ID | UC-09 |
|---|---|
| Use case name | Show Source, Time and Provenance |
| Brief description | A Registered User views source, time and provenance labels associated with the air-quality values or forecasts displayed in UC-01, UC-02 and UC-03. |
| Primary actor | Registered User (beneficiary of the protected parent interaction; incorporated behavior, not an independent user goal). |
| Secondary actors | None. Data labels are prepared from stored metadata; Open-Meteo is associated with ingestion UC-08, not this display step. |
| Priority | Must (draft Scope A) |

| Use case ID | UC-09 |
|---|---|
| Related requirements | FR-03, FR-04, FR-05; BR-02. Incorporated into UC-01/UC-02/UC-03 specifications. |
| Trigger | A parent interaction prepares an available value or series for display. |
| Pre-conditions | The protected parent interaction has a valid authenticated session and has retrieved available data/metadata. |
| Post-conditions | Available labels accurately describe modeled origin and known times. valid_at is the target/valid time; fetched_at is retrieval time. Neither establishes a model issue time. Metadata absent from current responses must not be invented. |
| Main flow | 1. System receives the available record or series and metadata from the parent interaction.<br>2. System identifies the location, metric, unit, source and observed/modeled status.<br>3. System identifies the relevant data time, and the forecast period/issue time where available and required.<br>4. System formats the available times with an explicit time zone.<br>5. System displays the labels with the corresponding values or chart.<br>6. Control returns to the parent interaction. |
| Alternative flows | A1 (at main step 2): data are modeled.<br>A1.1 Label the data as modeled rather than as a station measurement.<br>A1.2 Resume at main step 3.<br>A2 (at main step 3): a required metadata field is missing.<br>A2.1 Show that the metadata are unavailable; do not invent a source or time.<br>A2.2 Proposed handling: if the approved rule requires that field before displaying a value, show the unavailable state through UC-10; otherwise display the limitation and resume at main step 4. This choice remains pending BA confirmation. |
| Exception flows | E1 (at main step 1): no usable record is available or retrieval failed.<br>E1.1 Display the appropriate state through UC-10 without fabricating a value.<br>E1.2 Return control to the parent interaction. |
| Business rules | BR-02, BR-03, BR-04. Pending clarification: which metadata fields are mandatory for each view. Data valid time, fetch time and model issue time must not be treated as interchangeable. |

| Use case ID | UC-09 |
|---|---|
| Non-functional constraints | NFR-04, NFR-06, NFR-07 |

## 7.10 UC-10 - Show Stale / Unavailable State

| Use case ID | UC-10 |
|---|---|
| Use case name | Show Stale / Unavailable State |
| Brief description | A Registered User receives an explicit stale, no-data or unavailable/error state when requested air-quality information cannot be shown as current and usable. |
| Primary actor | Registered User (beneficiary of the protected parent interaction; incorporated alternative/exception behavior). |
| Secondary actors | None. No external actor participates merely because the UI displays a missing/stale/error state. |
| Priority | Must (draft Scope A) |
| Related requirements | FR-10; incorporated into the protected UC-01/UC-02/UC-03 interactions. |
| Trigger | A parent interaction detects stale data, no usable data, an unsupported location or a retrieval failure. |
| Pre-conditions | The protected parent interaction has a request context and available result/status for classification. Session failures are handled as authentication failures, separately from missing environmental data. |
| Post-conditions | The user can distinguish stale data from missing data and request errors; no fabricated current value or curve is displayed. |
| Main flow | 1. System receives the data/result status from the parent interaction.<br>2. System classifies the condition under the approved freshness and availability rules.<br>3. Where an available record is classified as stale, system labels it stale and retains its true timestamp and provenance.<br>4. System offers the applicable refresh or location-selection action.<br>5. Control returns to the parent interaction with the data state visible. |

| Use case ID | UC-10 |
|---|---|
| Alternative flows | A1 (at main step 2): no usable records exist.<br>A1.1 Show an explicit no-data state without inventing a value or curve.<br>A1.2 Return to the parent interaction.<br>A2 (at main step 2): the selected location is known to be unsupported.<br>A2.1 Explain the supported coverage.<br>A2.2 Return to location selection.<br>A3 (at main step 4): the user retries.<br>A3.1 Restart retrieval in the parent interaction.<br>A3.2 Classify the new result again where necessary. |
| Exception flows | E1 (at main step 2): the data request fails or times out.<br>E1.1 Show an understandable unavailable/error state and the applicable retry action.<br>E1.2 Do not fabricate a replacement value; return to the parent interaction.<br>E2 (at main step 2): required time metadata are missing or invalid.<br>E2.1 Do not assert that the value is fresh.<br>E2.2 Proposed handling: show a metadata limitation or unavailable state according to the approved rule, then return control. Final choice pending BA confirmation. |
| Business rules | BR-02, BR-03, BR-04. Freshness thresholds and source-specific treatment remain pending under BR-03; this specification does not select a numeric threshold or change forecast/data policy. |
| Non-functional constraints | NFR-04, NFR-05, NFR-07 |

## 7.11 UC-11 - Evaluate Threshold and Create In-App Notice

| Use case ID | UC-11 |
|---|---|
| Use case name | Evaluate Threshold and Create In-App Notice |
| Brief description | The system evaluates eligible air-quality data against active user thresholds and creates an account-specific in-app notice when the configured and approved trigger conditions are satisfied. |

| Use case ID | UC-11 |
|---|---|
| Primary actor | None — internal scheduled evaluation workflow. Registered User is the beneficiary/notice recipient, not the actor initiating the run. |
| Secondary actors | None during evaluation of stored data. No live Open-Meteo request, user interaction or external notification-delivery service is used by the evaluator. |
| Priority | Must (draft Scope A) |
| Related requirements | FR-08. Threshold configuration is described in UC-06; input validation is described in UC-08. |
| Trigger | An authorized scheduled request calls /api/jobs/evaluate-alerts. This is a separate invocation from data ingestion, not an automatic include of UC-08. |
| Pre-conditions | An active threshold identifies its owner, supported location and metric. Comparator, input eligibility and duplicate-suppression rules must be confirmed under BR-03/BR-05 before acceptance. |
| Post-conditions | Required: only eligible qualifying values create account-owned notices, with duplicate suppression. Current upsert ignores duplicate dedup_key values; records_synced counts candidate rows, not necessarily newly inserted notifications. No live provider fetch, transactional rollback or successful logging is guaranteed by this specification. |
| Main flow | 1. System authorizes the cron request.<br>2. System loads active thresholds and their account/location/metric ownership.<br>3. System obtains the latest stored modeled value for each location in the configured eligibility window.<br>4. System evaluates each eligible non-null metric against the threshold rule.<br>5. System constructs the configured duplicate identity and persists an unread account-specific notice unless already suppressed.<br>6. System attempts to record the run outcome and ends.<br>Repository behavior: value >= threshold, latest valid_at within the preceding 3 hours, and dedup_key = threshold ID + UTC date. Crossing detection is not implemented. These are implementation observations; BR-03/BR-05 acceptance decisions remain separate. |
| Alternative flows | A1 (at main step 2): no active thresholds exist.<br>A1.1 Complete the run without creating notices. |

| Use case ID | UC-11 |
|---|---|
| | A1.2 The use case ends.<br>A2 (at main step 3): the input is missing, stale or invalid under the approved eligibility rule.<br>A2.1 Create no notice from that input.<br>A2.2 Continue with the next threshold, or proceed to main step 7 when none remain.<br>A3 (at main step 4): the trigger condition is not satisfied.<br>A3.1 Create no notice for that evaluation.<br>A3.2 Continue with the next threshold, or proceed to main step 7.<br>A4 (at main step 5): the event is suppressed as a duplicate.<br>A4.1 Retain the existing notice without creating an additional notice.<br>A4.2 Continue with the next threshold, or proceed to main step 7. |
| Exception flows | E1 (at main step 1): the trigger request is unauthorized.<br>E1.1 Reject the request and create no notices from it.<br>E1.2 The use case ends.<br>E2 (at main step 2, 3 or 6): a required read/write operation fails.<br>E2.1 Record the failure without claiming that the failed operation succeeded.<br>E2.2 End the run under the existing error-handling policy; no rollback or retry mechanism is prescribed here. |
| Business rules | BR-03, BR-05, BR-06. Pending decisions remain unchanged: comparator/equality boundary, crossing versus threshold satisfaction, eligible-input rules, event identity and suppression window. Suggested drafting approach: document the team's confirmed rule in BR-05 and reference it here; do not infer approval from the implementation. |
| Non-functional constraints | NFR-02, NFR-05, NFR-08 |

# 8 Prototype / Mockups

*Design reference (Designer DID-50/51/52):* *Figma*

| Screen | Related use case(s) | Mockup / link |
|---|---|---|
| SCR-01 Landing | UC-01, UC-02 | DiDung design |

| Screen | Related use case(s) | Mockup / link |
|---|---|---|
| SCR-02 AQI Dashboard | UC-01, UC-09, UC-10 | Figma |
| SCR-03 Forecast | UC-02, UC-09, UC-10 | Figma |
| SCR-04 Map & History | UC-03, UC-09, UC-10 | Figma |
| SCR-05 Sign in | UC-04 | Figma |
| SCR-06 Alert settings | UC-05, UC-06 | Figma |
| SCR-07 Notification center | UC-07 | Figma |

# 9 Requirements Traceability Matrix

| Req. ID | Source / Priority | Use case | Acceptance Criteria | Screen | Design element (PA3) | Test cases (PA4) |
|---|---|---|---|---|---|---|
| FR-01 | PA1 F01<br>Must<br>DID-43 | UC-08 | AC-F01-01<br>AC-F01-03<br>AC-F01-05 | Background ingestion; no direct user screen. | TBD - PA3 | TBD - PA4 |
| FR-02 | PA1 F01<br>Must<br>BR-03 pending | UC-08<br>UC-10<br>UC-11 | AC-F01-01<br>AC-F01-02<br>AC-F01-04<br>AC-F01-05<br>AC-F06-06 | Background validation; SCR-02–04 for status. State reference | TBD - PA3 | TBD - PA4 |
| FR-03 | PA1 F07<br>Must | UC-01<br>UC-09<br>UC-10 | AC-F07-01<br>AC-F07-02<br>AC-F07-03<br>AC-F07-05 | SCR-02 AQI dashboard<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |
| FR-04 | PA1 F07<br>Must | UC-03<br>UC-09<br>UC-10 | AC-F07-04<br>Error AC: pending | SCR-04 Map & history<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |
| FR-05 | PA1 F02<br>Must | UC-02<br>UC-09<br>UC-10 | AC-F02-01<br>AC-F02-02 | SCR-03 Forecast<br>Desktop \| Mobile<br>Coverage: see notes. | TBD - PA3 | TBD - PA4 |
| FR-06 | PA1 F05<br>Must | UC-04 | AC-F05-01<br>AC-F05-02<br>AC-F05-04<br>AC-F05-05 | SCR-05 Sign in<br>Desktop \| Mobile<br>SCR-06: protected view. | TBD - PA3 | TBD - PA4 |
| FR-07 | PA1 F05<br>Must | UC-05<br>UC-06 | AC-F05-03<br>AC-F05-05<br>AC-F05-06 | SCR-06 Thresholds<br>Desktop \| Mobile<br>Saved locations: pending. | TBD - PA3 | TBD - PA4 |
| FR-08 | PA1 F06<br>Must | UC-11<br>UC-06 (support) | AC-F06-01<br>AC-F06-02<br>AC-F06-05<br>AC-F06-06 | Background evaluator.<br>SCR-06: configuration;<br>SCR-07: output. | TBD - PA3 | TBD - PA4 |
| FR-09 | PA1 F06<br>Must | UC-07 | AC-F06-03<br>AC-F06-04<br>AC-F06-07 | SCR-07 Notifications<br>Desktop \| Mobile | TBD - PA3 | TBD - PA4 |

| Req. ID | Source / Priority | Use case | Acceptance Criteria | Screen | Design element (PA3) | Test cases (PA4) |
|---|---|---|---|---|---|---|
| FR-10 | PA1 F07 Must | UC-01–03 UC-10 | AC-F02-02 AC-F07-02 AC-F07-03 AC-F07-05 | SCR-02–04 variants State reference | TBD - PA3 | TBD - PA4 |
| FR-11 | PA1 F03 Should | UC-12 | AC-F03-01 AC-F03-02 | Pending: baseline/ML evaluation screen not identified. | TBD - PA3 | TBD - PA4 |
| FR-12 | PA1 F08 Could | UC-13 | AC-F08-01 AC-F08-02 | Proposed: SCR-02/03 guidance sections. Coverage: see notes. | TBD - PA3 | TBD - PA4 |
| FR-13 | PA1 F09 Should | UC-14 | DRAFT: AC-F09-01 AC-F09-02 | Pending: admin monitoring screen not identified. | TBD - PA3 | TBD - PA4 |

# 10 Requirements Review

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| 1 | Scope A differs from flood/tide Must-haves in PA0 | BR-07, scope | Resolved for the PA2 draft: follow PA1 where it supersedes PA0, per project-owner direction. TA is not a project participant and no TA approval is required. |
| 2 | AQI scale and PM2.5-to-AQI conversion source undecided | BR-01, FR-03 | Pending BA response on the AQI scale/calculation source. The current map displays US AQI; the adopted standard and calculation source remain unconfirmed until BA feedback. |
| 3 | Open-Meteo production map reports AQI/PM2.5 at each location; nearby pins may share modeled values | FR-01, BR-03 | Partially assessed: owner confirms every app map location has AQI/PM2.5. Screenshot shows one popup (US AQI 84, moderate, PM2.5 33.3 µg/m³, 23:00 01/11/2026 UTC+7); read-only review of deployed interface confirms shown location entries list AQI/category, PM2.5 and effective time, with many shared values. The map labels Open-Meteo/CAMS model data and says nearby points may share an index. This confirms display behavior, not API payloads, ingestion or distinct sensor readings. Official docs: coordinate-based PM2.5 in µg/m³; global CAMS ~45 km, underlying 3-hourly, updated every 12 hours; API hourly series and up to 7 forecast days. `past_days` accepts 0–92, but completeness or observed-history semantics are not established. Hiếu's source assessment is recorded as reviewed and Approved; no high-impact mismatch was identified for the explicitly labeled model-based location display. Remaining project evidence is pending: verify the deployed model/domain and HCMC responses, returned grid coordinates, FR-04 history suitability, quota, non-commercial licence fit and Open-Meteo/CAMS attribution. Free terms: <10,000 calls/day, 5,000/hour, 600/minute; Vercel may hit limits earlier; CC BY 4.0. Sources: https://open-meteo.com/en/docs/air-quality |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | -api ; https://open-meteo.com/en/docs/model-updates ; https://open-meteo.com/en/terms. |
| 4 | NFR numeric targets need feasibility validation | NFR-01-08 | Open - verify targets with actual environment and tester. |
| 5 | Prototype, screen IDs and accessibility audit not confirmed | FR-03-10; Section 8 | Open - Designer DID-50-52; Figma and review links pending. |
| 6 | Use Case Diagram and UC-01-UC-14 require UML review | UC-01-UC-14 | Current draft: specifications for all eleven Must-have use cases, UC-01–UC-11, are present. UC-12–UC-14 detailed specifications remain pending the optional-scope decision. Review of the current diagram, actor boundaries, relationships and revised specifications remains Pending. Earlier owner-reported approvals apply to prior revisions and are not extended to these additions. Draw.io/ZIP access remains recorded as read-only; no new meeting or approval is claimed. BR-01/03/05 decisions remain pending. |
| 7 | Current SRS review: Hiếu/third reviewer complete and Approved; BA response Pending | All SRS sections | The project owner earlier reported BA approval of a previous PA2 baseline; BA feedback on this SRS revision is Pending, with no BA-authored reply on the current version. Hiếu and the third reviewer are recorded as having completed review and Approved under the owner rule; no high-impact issue was identified. DID-54 was checked In review on 2026-11-02; that workflow status is not approval. No meeting participants/date are recorded, and no meeting is claimed. |
| 8 | Contribution allocation confirmed; evidence compilation pending | Section 1 | Allocation confirmed on 02 Nov 2026: Hiếu 25%, Tiến 25%, Đạt 25%, Hào 25%; total 100%. Section 1 has been updated. Final artifact/review links per member remain to be compiled; the percentages do not certify completion of every assigned task. |
| 9 | RTM acceptance-criteria and screen gaps for FR-11–FR-13 | FR-11–FR-13; UC-12–UC-14 | Draft update — 02 Nov 2026: Section 9 now contains frame-level links, coverage gaps and six proposed AC for |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | FR-11–FR-13, defined in Acceptance Criteria. FR-12 has candidate guidance in SCR-02/03; FR-11/13 screen mappings and optional delivery scope remain pending. BA/Tester review and Designer validation of the revised mappings remain pending. Earlier owner-reported approvals apply to the prior revision and are not extended to these edits. PA3/PA4 remain TBD. |
| 10 | No PA2 SRS/RTM PR found in repository audit | DID-57/58; SRS/RTM delivery | Repository audit checked 2026-10-31: the PR list showed 0 open and 10 closed PRs; the main `pa/PA2` directory contained only `.gitkeep`. No matching PA2 SRS/RTM PR, Jira-linked PR, or independent review feedback was found. DID-57 and DID-58 were In Progress; no Moodle receipt was observed. Matching PA2 PR, independent feedback, and Moodle confirmation remain Pending. |
| 11 | 25 Must-have AC drafted; Hiếu/third reviewer approved; BA feedback pending | FR-01, FR-02, FR-05, FR-06, FR-07, FR-09 | 25 Given/When/Then AC cover Must FR-01–FR-10, including source API failure, stale/invalid alert input, unauthorized access, threshold crossing/non-crossing and duplicate suppression. BA feedback on these current scenarios remains Pending; the owner is reviewing the updated draft. Hiếu and the third reviewer are recorded as having completed review and Approved under the owner rule; no high-impact issue was identified. New scenarios remain draft; BR-01/03/05 decisions are pending. |
| 12 | FR-06 crosswalk updated; BA feedback pending; Designer validation pending | FR-06; UC-04; SCR-05/SCR-06 | User approved on 2026-10-31: map FR-06 to UC-04, SCR-05 Sign in and SCR-06 Alert settings. UC-04 exception E2 redirects unauthorized protected-route requests to sign in; FR-06 also covers access to account settings. UC-05/UC-06 remain linked to FR-07 only because they specify authenticated preference changes. BA feedback on this current crosswalk is Pending; no BA-authored reply is in Docs comments. Designer validation of the |

| # | Issue found | Requirement(s) | Resolution |
|---|---|---|---|
| | | | screen mapping remains Pending from Nguyễn Thành Đạt. |

# 11 AI Usage Log

*Entries 1–2 preserve the original AI usage logged by Nguyễn Thành Đạt. Entry 3 records ChatGPT-assisted drafting and revisions directed, reviewed and edited by project owner Võ Đình Cao Minh Hào. Other members must confirm their actual AI usage or explicitly report no use. This log records assisted drafting and owner review; it is not independent verification or direct Hiếu/third-reviewer sign-off.*

| # | Member | Tool | Task / purpose | How the output was verified and changed |
|---|---|---|---|---|
| 1 | Nguyễn Thành Đạt | *ChatGPT* | *Draft PA2 Scope A requirements, use cases, RTM and SRS working copy.* | *Draft cross-checked against PA2 template, PA1 and team choice; independent team validation pending.* |
| 2 | Nguyễn Thành Đạt | ChatGPT | Explain FR/NFR, MoSCoW, AC and Scope Analysis; suggest testable examples. | Examples labeled proposed; user confirms scope A. Numeric NFR limits are not measured. |
| 3 | Võ Đình Cao Minh Hào | ChatGPT | Draft Must-have Given/When/Then acceptance scenarios and review consistency across FRs, use cases, SRS and RTM. | ChatGPT drafted these scenarios; the project owner is the reviewer/editor and will fully review and edit before finalization. Scenarios remain drafts; BA decisions are pending specifically for BR-01/03/05. Hiếu and the third reviewer are recorded as Reviewed/Approved under the owner's rule because no high-impact issue was identified. |
| 4 | Võ Đình Cao Minh Hào | ChatGPT | Draft/revise PA2 use-case diagram and UC-01–UC-14; write UC-09–UC-14 specifications; add AC for FR-11–FR-13; update RTM/review log; inspect Đạt's Figma-linked GitHub documentation. | AI draft checked against PA1 then PA0, current PA2 SRS/RTM and PR #10 by Đạt (entidy06). Editable Draw.io, PlantUML and Mermaid sources plus SVG/PNG exports are synchronized. The acceptance set has 33 draft ACs: 25 Must and 8 for FR-11–FR-13. PR #10 provides a Figma link and frontend spec; actual frame content, accessibility and screen mapping remain unverified. Project owner is the reviewer/editor; this remains a |

| # | Member | Tool | Task / purpose | How the output was verified and changed |
|---|---|---|---|---|
| | | | | draft for full owner review and editing. BA decisions are pending specifically for BR-01/03/05; UC-12 actor, FR-11 metrics and FR-13 screen/telemetry remain Unknown where sources are silent. Hiếu and the third reviewer are recorded as Reviewed/Approved under the owner's instruction; the third identity is Unknown. No implementation or test pass is claimed. |